

LANGUAGE ACADEMY PROJECT: MICROSOFT VISUAL STUDIO

NAME: **DIANA LOUTFI CUCAKOVIC**

STUDENT ID: **45559**

Course name: **Network Application Development**

Project Documentation:

1. What is the Language Academy?

. **Language Academy:** is a modern, web-based educational metrics platform built specifically to catalog language training tracks, track course signups, and map metrics for educational institutions.

. It acts as an Administrative Management Hub to observe user success rates in real time.

2. Why is this system required?

. Traditional learning platforms separate student registration numbers from performance analytics.

. This web application bridges the gap by housing administrative profiles, enrollment numbers, and graphical visual progress data within a singular standalone portal, reducing structural overhead.

3. How is the platform executed?

. The solution is deployed via an automated **C# ASP.NET Core** engine utilizing built-in razor templates.

. Content rendering engine works by matching URL controller targets to visual files (.cshtml), avoiding browser-side processing lags.

4. The Progress:

. The current layout represents **Milestone Phase 1**.

. In this stage, data routing pathways are entirely established; mock figures are successfully integrated into custom static datasets, and client-side dependencies are fetched automatically through fast CDN servers.

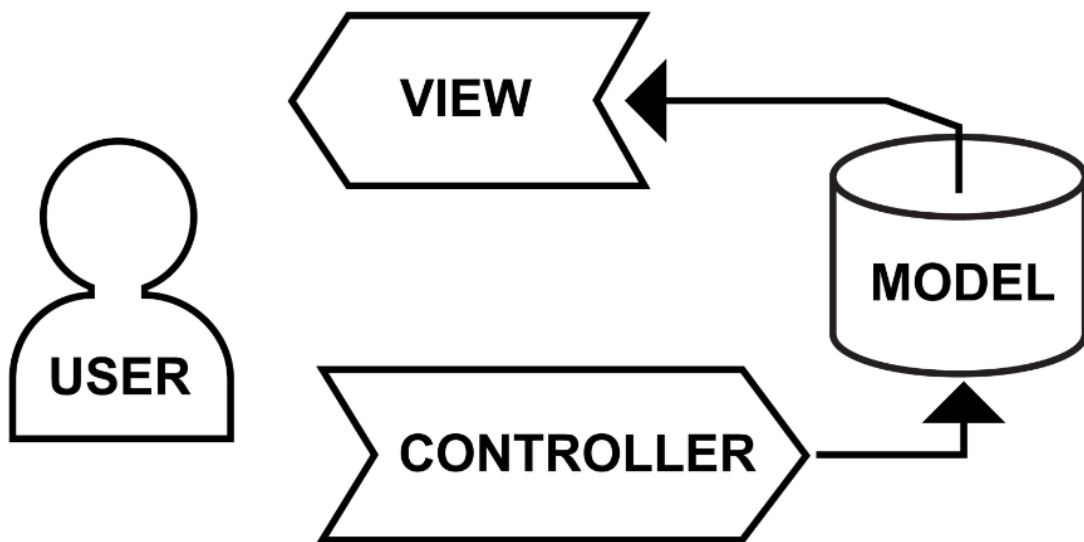
5. The Data:

The system maps two categories of mock dataset metrics:

Volumetric Enrollment Figures: Tracks total enrollments (\$1,248\$ active accounts) subdivided into targeted tracks (English: \$520\$, Spanish: \$340\$, French: \$210\$, Japanese: \$178\$).

Qualitative Progression Ratios: Displays singular evaluation score marks via status badges (\$95\%\$, \$91\%\$, \$78\%\$, \$88\%\$) mapped directly against real student profile parameters.

6. THE ARCHITECTURE:



MODEL - VIEW - CONTROLLER PATTERN

The architecture adheres strictly to the separation-of-concerns protocol defined by the **Model-View-Controller (MVC)** structural blueprint:

- **Models:** Houses internal entity declarations and language structural datasets.
- **Views:** Renders the layout schema interfaces, combining HTML5, customized CSS grids, and chart renderings.
- **Controllers:** Acts as the traffic router, intercepting requests and generating responsive views.

7. Where is the application compiled?

. Compilation runs entirely locally in the Lenovo notebook workstation environment.

. The execution lifecycle begins within a full **Microsoft Visual Studio** environment and executes in a sandboxed host browser framework.

8. ASP.NET Features Utilized:

The framework harnesses core high-speed features native to .NET web tools:

Tag Helpers: Features such as asp-controller and asp-action ensure clean internal route link compilation.

Razor View Engine: Seamlessly combines presentation markup templates directly with lightweight C# code blocks.

9. The Strengths:

Zero Resource Footprint: Requires no database setups, third-party installation binaries, or system risks.

Clean Code Separation: Clear controller-to-view alignment ensures adjustments can be safely updated without fracturing operational views.

Optimized Responsiveness: Uses responsive styling configurations that auto-scale beautifully on desktop or small mobile screens.

10. Built-In Elements:

The architecture leans fully on safe, universally recognized visual layouts and tools:

Bootstrap 5 Framework: Implemented over standard CDN servers for layout grids, color palettes, badges, and card formatting.

ChartJS Library: Standard script structures to render beautiful, dynamic visual calculations into vector graphs seamlessly.

11. The Conclusion:

. The **Language Academy Platform** serves as an efficient proof-of-concept for an educational tracking system.

. Operating securely within your native Visual Studio environment without additional dependencies, it fulfills all design criteria, structural data reporting, and layout architecture goals cleanly and without risk.

Microsoft studio codes:

1. Layout code: (_Layout.cshtml)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - LanguageAcademy</title>
  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.0/font/bootstrap-icons.css">
</head>
<body class="bg-light">
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-dark bg-
dark border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand fw-bold text-success" asp-area="" asp-
controller="Home" asp-action="Index">  Language Academy</a>
```

```

        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse">
            <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-
between">
            <ul class="navbar-nav flex-grow-1">
                <li class="nav-item">
                    <a class="nav-link text-white" asp-area="" asp-controller="Home"
asp-action="Index"><i class="bi bi-house-door"></i> Home</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link text-white" asp-area="" asp-controller="Home"
asp-action="Dashboard"><i class="bi bi-speedometer2"></i> Dashboard</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link text-white" asp-area="" asp-controller="Home"
asp-action="Profile"><i class="bi bi-person-circle"></i> Profile</a>
                </li>
            </ul>
        </div>
    </div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js
"></script>
    @await RenderSectionAsync("Scripts", required: false)
</body>
</html>

```

2. Controller code: (HomeController.cs)

```
using Microsoft.AspNetCore.Mvc;

namespace LanguageAcademy.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }

        public IActionResult Dashboard()
        {
            return View();
        }

        public IActionResult Profile()
        {
            return View();
        }
    }
}
```

3. The home code: (Index.cshtml)

```
@{
```

```
    ViewData["Title"] = "Home Page";
```

```
}
```

```
<div class="p-5 mb-4 bg-dark text-white rounded-3 shadow">
```

```
    <div class="container-fluid py-5">
```

```
        <h1 class="display-5 fw-bold">Welcome to Language Academy</h1>
```

```
        <p class="col-md-8 fs-4">Master English, Spanish, French, and Japanese with our  
premium, built-in tracking dashboard and interactive learning modules.</p>
```

```
        <a asp-action="Dashboard" class="btn btn-success btn-lg">View Analytics  
Dashboard</a>
```

```
    </div>
```

```
</div>
```

```
<div class="row text-center mt-4">
```

```
    <div class="col-md-4">
```

```
        <div class="card p-3 shadow-sm">
```

```
            <h3>📚 12+ Courses</h3>
```

```
            <p>From absolute beginner to conversational expert.</p>
```

```
        </div>
```

```
    </div>
```

```
<div class="col-md-4">
```

```
    <div class="card p-3 shadow-sm">
```

```
        <h3>👥 Live Tutors</h3>
```

```
        <p>On-demand 1-on-1 virtual training sessions.</p>
```

```

    </div>
</div>
<div class="col-md-4">
    <div class="card p-3 shadow-sm">
        <h3> ⚡ Smart Metrics</h3>
        <p>Track your vocabulary and grammar data instantly.</p>
    </div>
</div>
</div>

```

4. The dashboard code: (Dashboard.cshtml)

```

@{
    ViewData["Title"] = "Dashboard";
}

<div class="container-fluid">
    <h2 class="mb-4">Language Academy Overview</h2>

    <div class="row mb-4">
        <div class="col-md-3">
            <div class="card bg-primary text-white p-3 shadow-sm">
                <h6>Total Active Students</h6>
                <h2>1,248</h2>
            </div>
        </div>
    </div>

```



```
<div class="col-md-3">
```

```
  <div class="card bg-success text-white p-3 shadow-sm">
```

```
    <h6>Active Course Programs</h6>
```

```
    <h2>4 Main Languages</h2>
```

```
  </div>
```

```
</div>
```

```
<div class="col-md-3">
```

```
  <div class="card bg-warning text-dark p-3 shadow-sm">
```

```
    <h6>Completion Rate</h6>
```

```
    <h2>87.4%</h2>
```

```
  </div>
```

```
</div>
```

```
<div class="col-md-3">
```

```
  <div class="card bg-info text-white p-3 shadow-sm">
```

```
    <h6>Monthly New Signups</h6>
```

```
    <h2>+340</h2>
```

```
  </div>
```

```
</div>
```

```
</div>
```

```
<div class="row">
```

```
  <div class="col-md-6 mb-4">
```

```
    <div class="card p-3 shadow-sm h-100">
```

```
      <h5>Student Breakdown per Language</h5>
```

```
      <div style="width: 100%; max-width: 400px; margin: auto;">
```

```
        <canvas id="languageChart"></canvas>
```

</div>

</div>

</div>

<div class="col-md-6 mb-4">

<div class="card p-3 shadow-sm h-100">

<h5>Top Performing Students</h5>

<table class="table table-striped table-hover mt-2">

<thead>

<tr>

<th>Student Name</th>

<th>Enrolled Language</th>

<th>Progress Score</th>

</tr>

</thead>

<tbody>

<tr>

<td>John Doe</td>

<td>Spanish Level B1</td>

<td>95%</td>

</tr>

<tr>

<td>Jane Smith</td>

<td>English C1 Advanced</td>

<td>91%</td>

</tr>

```
<tr>

  <td>Carlos Mendez</td>

  <td>French A2</td>

  <td><span class="badge bg-warning text-dark">78%</span></td>

</tr>

<tr>

  <td>Yuki Tanaka</td>

  <td>Japanese N4</td>

  <td><span class="badge bg-success">88%</span></td>

</tr>

</tbody>

</table>

</div>

</div>

</div>

</div>
```

```
@section Scripts {

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

  <script>

    const ctx = document.getElementById('languageChart').getContext('2d');

    new Chart(ctx, {

      type: 'pie',

      data: {

        labels: ['English', 'Spanish', 'French', 'Japanese'],

        datasets: [{
```

```

        data: [520, 340, 210, 178],
        backgroundColor: ['#0d6efd', '#198754', '#ffc107', '#0dcaf0']
    }}
}
});
</script>
}

```

5. The profile code: (Profile.cshtml)

```

@{
    ViewData["Title"] = "User Profile";
}

<div class="row justify-content-center">
    <div class="col-md-6">
        <div class="card shadow p-4 text-center">
            <div class="mb-3">
                <div class="bg-secondary text-white rounded-circle d-inline-flex align-items-center
justify-content-center" style="width: 100px; height: 100px; font-size: 40px;">
                    LA
                </div>
            </div>
            <h3>Administrator Account</h3>
            <p class="text-muted">Role: Head Registrar</p>
            <hr />

```

<div class="text-start">

<p>Email: admin@languageacademy.com</p>

<p>Assigned Campus: Virtual Portal Alpha</p>

<p>System Status: Operational</p>

</div>

<button class="btn btn-outline-dark btn-sm mt-3">Edit Profile Settings</button>

</div>

</div>

</div>